AI-Based Contract Risk Analysis and Clause Recommendation System

PHASE I REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING





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December 2026

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled "AI-BASED CONTRACT RISK ANALYSIS AND CLAUSE RECOMMENDATION SYSTEM" is the bonafide work of **NITHIESH S (230701215), MUHAMMED NAVEED KHAN M J (230701200), PRANAV SRIRAM (230701235), PRANESH SENTHILKUMAR (230701236)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

"This project addresses the following Sustainable Development Goal(s): Goal – SDG 16 : Peace, Justice and Strong Institutions."

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ABSTRACT

Contracts form the basis of almost all commercial relationships, but the manual examination of long legal documents is still a slow, expensive process and is prone to human error. Clauses that pose a risk—such as unlimited liability, automatic renewal, one-sided indemnity, or unclear terms regarding termination and payment—can lead organisations to face serious legal and financial repercussions if they are not identified. Currently available automated tools are mainly limited to keyword searching or clause-level classification, provide little scope for organisation-specific customisation, and seldom explain their decisions or suggest revised wording.

This paper outlines the design of an AI-Based Contract Risk Analysis and Clause Recommendation System which takes PDFs and Word documents as input, extracts and divides the clauses, classifies them into standard legal categories, assigns a risk severity on a four-level scale (Low, Medium, High, Critical), and uses a locally hosted Large Language Model served through Ollama to generate clause recommendations that are grounded in the context of the contract and organisational guidelines.

The proposed architecture integrates transformer-based clause classification (using LegalBERT/RoBERTa/DeBERTa models fine-tuned on the CUAD dataset) with a locally hosted Large Language Model served through Ollama for generating the recommendations, all made accessible via an interactive dashboard that provides risk visualisation, explanations linked to the source material, and exportable reports. The design is inspired by recent research into LLM-based legal risk assessment using Ollama and extends that work towards an end-to-end, explainable, organisation-adaptable process covering the entire pipeline from ingestion to recommendation.

This Phase I report details the present state of the prototype implementation, the modules that are still proposed or planned, and a credible evaluation methodology that will be employed to assess the system when it is fully realised.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman Mr. S. MEGANATHAN, B.E, F.I.E., our Vice Chairman Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S., and our respected Chairperson Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D., for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to Dr. S.N. MURUGESAN, M.E., Ph.D., our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to Dr. J. Manoranjini, Ph.D., Professor and Head of the Department of Computer Science and Engineering for her guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, Dr. N. Srinivasan, Professor, Department of Computer Science and Engineering for his valuable guidance throughout the course of the project.

We thank our Project Coordinator, Mr. Karthik, B.E., M.E., Assistant Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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1. INTRODUCTION

1.1 GENERAL

Contracts form the backbone of nearly every business transaction, defining the rights, obligations and risk allocation between parties. Because a contract's wording ultimately determines an organisation's exposure to financial risk, careful review before signing is not optional but essential. As the volume of commercial agreements handled by modern organisations grows, the process of reviewing them by hand struggles to keep pace.

Manual contract review is time-consuming, expensive and difficult to scale. A single agreement may run to dozens of pages of dense legal language, and identifying the handful of clauses that carry material risk requires sustained expert attention. This dependence on human reviewers introduces inconsistency: two lawyers examining the same clause may reach different conclusions about its severity, and fatigue or time pressure can cause important risk indicators to be overlooked entirely.

The consequences of missing or poorly worded clauses can be severe. An unlimited liability clause can expose a company to damages far beyond the value of the underlying contract. An automatic renewal clause with a narrow cancellation window can lock an organisation into unfavourable terms for years. A one-sided indemnity clause can shift disproportionate risk onto one party. Ambiguous termination or payment conditions can create prolonged and costly disputes. Because such clauses are often embedded in routine boilerplate language, they are easy to overlook during a fast manual read-through.

In parallel, the past decade has seen substantial progress in Natural Language Processing, particularly with the emergence of Transformer-based architectures, BERT-family encoder models and Large Language Models. These models have demonstrated a strong ability to understand and generate natural language. More recently, local Large Language Model via Ollama has enabled such models to ground their outputs in external domain-specific knowledge rather than relying solely on static pre-trained parameters. These advances make it increasingly feasible to automate portions of contract analysis that were previously restricted to human experts.

This project aims to design and develop an intelligent system that reduces the burden of manual contract review while improving consistency, accuracy and decision-making quality by combining modern NLP techniques with organisation-specific knowledge.

1.2 OBJECTIVE

The primary objective of this project is to design and develop an AI-Based Contract Risk Analysis and Clause Recommendation System that automatically processes legal contracts and assists legal and business teams in identifying risks and improving clause language.

The specific objectives of the system are as follows:

- To automatically extract text from contract documents in PDF and DOCX formats, including scanned documents using Optical Character Recognition (OCR).
- To segment the extracted text into meaningful individual clauses that each address a single contractual topic.
- To classify each clause into appropriate legal categories using transformer-based models such as LegalBERT, RoBERTa or DeBERTa fine-tuned on the CUAD dataset.

- To assign a risk severity level (Low, Medium, High or Critical) to every clause based on classification results, high-risk patterns and similarity to known risky formulations.
- To generate improved clause recommendations using a locally hosted Large Language Model served through Ollama, guided by organisational guidelines and contract context.
- To generate improved and safer clause recommendations using a Large Language Model that is grounded in the retrieved organisational context.
- To present the analysis results through an interactive dashboard that provides risk visualisation, source-linked explanations and exportable reports.
- To reduce the time and effort required for manual contract review while improving consistency, transparency and organisational adaptability.

1.3 EXISTING SYSTEM

Traditional contract review remains largely a manual process. Legal teams read entire documents, highlight risky clauses and suggest modifications based on personal experience and knowledge of company policies. Some organisations employ basic rule-based tools or simple keyword-search systems that look for predefined risky terms such as “unlimited liability” or “automatic renewal”.

These approaches suffer from several important limitations. Rule-based and keyword-driven systems fail to generalise to paraphrased or novel clause wording. Academic systems built around BERT-style classifiers typically stop at labelling a clause’s category or its risk level without proposing an improved alternative. Commercial Contract Lifecycle Management (CLM) platforms are largely closed, expensive and not

easily customisable to a given organisation's playbooks and risk appetite.

Across both academic and commercial systems, explainability is frequently weak: end users are shown a verdict without a clear evidence-linked justification, which undermines trust and adoption. The dependence on human expertise also leads to inconsistency across different reviewers, and the process does not scale well when the volume of contracts increases. Existing tools have limited ability to learn from past agreements and internal organisational policies.

This gap—between systems that merely classify or score clauses and the practical need for an organisation-adaptable end-to-end review assistant that also recommends improved wording—motivates the present work.

1.4 PROPOSED SYSTEM

The proposed AI-Based Contract Risk Analysis and Clause Recommendation System is designed to close the identified gap by combining clause-level classification, multi-level risk scoring and Ollama-based recommendation generation grounded in the contract context and organisational guidelines, all surfaced through an explainable dashboard-driven interface.

The principal contributions of this work are: (i) an end-to-end pipeline spanning document ingestion, clause segmentation, classification and risk scoring through to recommendation generation; (ii) the use of a locally hosted Ollama LLM to generate clause recommendations that are guided by organisational guidelines and the surrounding contract context; and (iii) an explainability layer that links every risk verdict and recommendation to its supporting evidence.

The system accepts PDF and Word documents as input. Text is extracted using pdfplumber, with Tesseract OCR as a fallback for scanned pages. The extracted text is segmented into individual clauses. Each clause is classified by a transformer model fine-tuned on the CUAD dataset. A risk scoring module assigns one of four severity levels. For high and critical clauses, the original clause together with relevant organisational guidelines and surrounding contract context is passed to a locally hosted Large Language Model served through Ollama that generates safer alternative wording. Results are presented on an interactive dashboard with visual risk summaries and downloadable reports.

2. LITERATURE SURVEY

2.1 LITERATURE REVIEW

Contract analysis has been an active research area in LegalTech and Natural Language Processing for several years. Early systems relied mainly on rule-based methods and keyword matching. With the development of deep learning and transformer models, the accuracy of clause classification and risk detection has improved significantly.

LEGAL-BERT (Chalkidis et al., 2020) is a domain-specific BERT model pre-trained on large legal corpora. It significantly outperforms general BERT models on legal text classification and named entity recognition tasks. This model has become a foundation for many subsequent legal NLP applications and is used as one of the candidate classification backbones in the present system.

The Contract Understanding Atticus Dataset (CUAD) introduced by Hendrycks et al. (2021) is a large expert-annotated dataset containing 41 different clause categories. It has become a standard benchmark for evaluating contract review systems. Many recent papers use CUAD for training and testing clause classification models, and the present work also leverages CUAD for fine-tuning and evaluation of the classification component.

local Large Language Model via Ollama, proposed by Lewis et al. (2020), combines the strengths of information retrieval and generative models. It allows a language model to access external knowledge bases, making the generated responses more accurate and grounded in real documents. Ollama is particularly useful for incorporating organisation-specific policies into recommendations and forms a core part of the proposed architecture.

Several studies have applied Natural Language Inference techniques to determine whether a contract clause entails, contradicts or is neutral with respect to a given statement. Other research works have explored the use of GPT-4 and similar large language models for clause generation, risk explanation and automated redlining. Moon et al. (2022) demonstrated effective detection of contractual risk clauses using BERT. Singh et al. (2025) surveyed classification tasks and approaches for legal contracts. Zhang et al. (2026) explored automated multi-label classification of risk clauses using GPT-driven augmentation.

Recent papers from 2023 to 2026 have explored multi-agent systems, knowledge-graph-enhanced Ollama pipelines and hybrid architectures that combine classification models with generative models and retrieval systems for more reliable contract analysis. Wang (2024) presented Rainbow Ollama, an LLM-powered Ollama system specifically for contract review. The Azbil research team (2025) proposed a legal contract document risk assessment method based on large language models that combines Ollama with false-negative reduction and dynamic window sizing. Vishwakarma et al. (2026) examined AI-based risk detection in Indian legal contracts using NLP and LLMs.

2.2 COMPARISON AND DISCUSSION

The reviewed literature provides strong foundations for individual components of contract analysis such as clause classification, semantic retrieval and generative recommendation. However, most existing systems focus either on classification or on generation, but rarely provide a complete end-to-end pipeline that includes risk scoring and policy-aware recommendation.

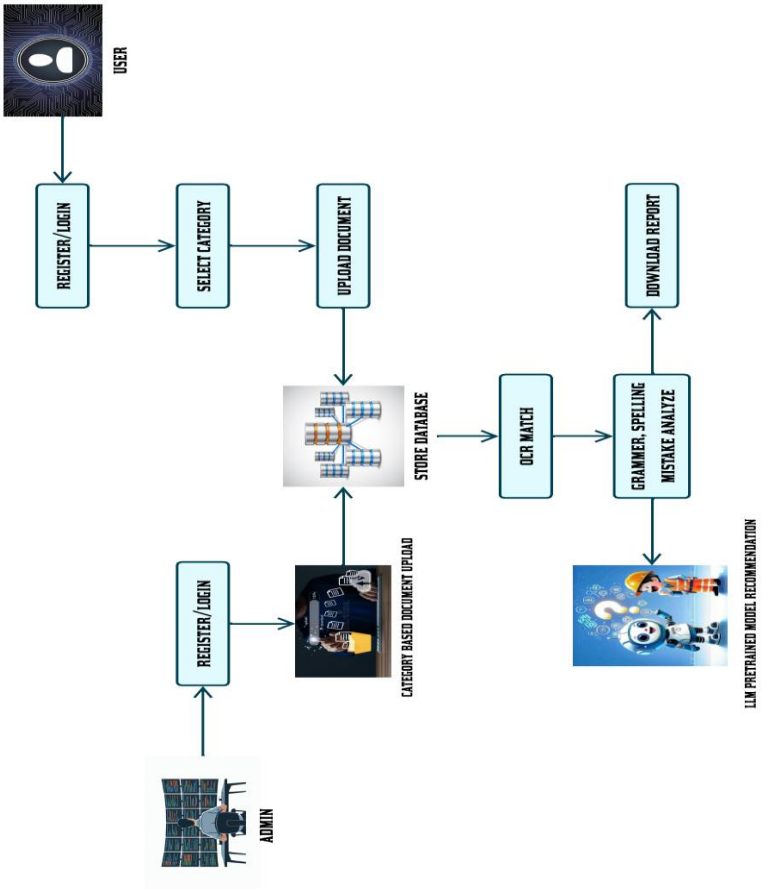
Limited ability to adapt to organisation-specific policies and historical agreements remains a significant gap. Many systems also lack sufficient explainability of the assigned risk scores and do not provide human-in-the-loop mechanisms for continuous improvement. High-performing commercial systems are often expensive and not openly available for academic experimentation.

The proposed system addresses these gaps by integrating clause classification (LegalBERT/RoBERTa/DeBERTa fine-tuned on CUAD), multi-level risk scoring, Ollama-based LLM recommendation guided by organisational guidelines into a single coherent pipeline. An explainability layer links every risk verdict and recommendation to supporting evidence. This combination aims to deliver both accuracy and practical usefulness for real-world contract review while remaining adaptable to different organisational risk appetites and playbooks.

3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The system follows a clear modular pipeline architecture that covers the complete journey from document upload to actionable recommendation:



Each stage is designed as an independent module. This modular design allows easy maintenance and future enhancement of individual components without affecting the entire system. The architecture ensures that the system can process both digital and scanned contracts and produce actionable insights for legal and business users.

The classification stage employs transformer-based models (LegalBERT, RoBERTa or DeBERTa) fine-tuned on the CUAD dataset. The recommendation stage uses a locally hosted Large Language Model served through Ollama that receives the original clause together with relevant organisational guidelines and context. An interactive dashboard surfaces risk visualisations, source-linked explanations and exportable reports.

The design is inspired by recent research into LLM-based legal risk assessment using Ollama and extends that work towards an end-to-end, explainable, organisation-adaptable process covering the entire pipeline from ingestion to recommendation.

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

Software Component	Suggested Version (or) Release	Purpose
Python	3.10 or higher	Used as the primary programming language for data processing, NLP, machine learning and system development.
Transformers (Hugging Face)	Latest stable	Used for loading and fine-tuning the LegalBERT, RoBERTa and DeBERTa classification models.
Ollama	Latest stable	Used for serving the locally hosted Large Language Model that generates clause recommendations.
Ollama client / prompt libraries	Latest stable	Used for prompt construction and for interaction with the local Ollama LLM service.
pdfplumber	Latest stable	Used for text extraction from digital PDF contracts.
python-docx	Latest stable	Used for extracting text and structure from DOCX contracts.
Tesseract OCR (via pytesseract)	Latest stable	Used as a fallback to extract text from scanned documents.
Streamlit / FastAPI	Latest stable	Used for the interactive user interface and the API layer.
ReportLab	Latest stable	Used for generating exportable PDF reports.
openpyxl	Latest stable	Used for generating exportable Excel reports.
Supporting libraries	Latest stable	Used for data handling, embedding generation and visualisation of risk results.

3.2.2 HARDWARE REQUIREMENTS

Hardware Component	Recommended Specification	Purpose
Processor (CPU)	Multi-core CPU	Used for document parsing, clause segmentation, risk scoring and running the application and API layer.
Memory (RAM)	At least 16 GB	Used for loading transformer models, handling document batches and running the local Ollama LLM service.
Graphics Processor (GPU)	CUDA-capable GPU (preferred)	Used for faster fine-tuning of classification models and faster inference of the local Large Language Model.
Storage	Several tens of GB (medium-sized organisation)	Used to store historical contracts, policy documents and generated reports in local storage; capacity depends on document volume.
Deployment environment	Cloud instance with similar or higher specifications	Used for hosting the system in production with the same or greater compute capacity as the development machine.

3.2.3 DATABASE REQUIREMENTS

Metadata about contracts, clauses, risk scores and recommendations can be stored in a conventional relational database (PostgreSQL or SQLite) or a lightweight document store for easy retrieval, auditing and reporting. The local Ollama service hosts the Large Language Model used for generating clause recommendations.

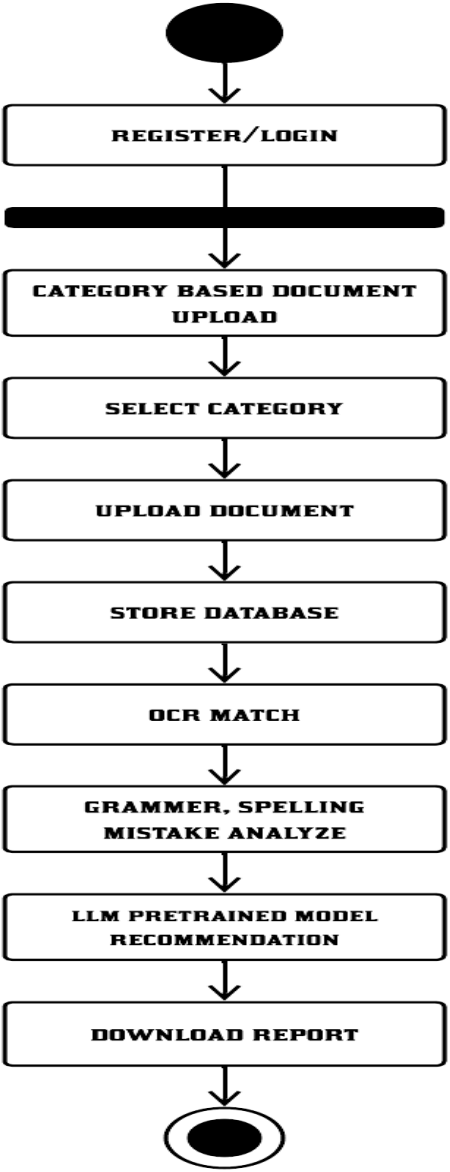
3.2.4 DEPLOYMENT AND SCALING

The system can be deployed as a web application accessible to authorised users within an organisation. Horizontal scaling can be achieved by separating the frontend, backend API, model inference service and local storage into independent services. Caching of frequent queries, asynchronous processing of large documents and batch embedding generation help maintain responsive performance under load.

3.3 SYSTEM DESIGN

3.3.1 ACTIVITY DIAGRAM

The activity diagram illustrates the complete flow of activities from the moment a user uploads a contract until the final report is generated. The main activities include document upload, text extraction (with an OCR branch for scanned pages), clause segmentation, classification, risk scoring, conditional retrieval of relevant policies (only for High and Critical clauses), generation of recommendations, and display of results on the dashboard. Decision points exist for handling scanned versus digital documents and for determining whether a clause requires recommendation generation based on its risk level. The flow also includes an optional human-in-the-loop review step before final acceptance of recommendations.


REGISTER/LOGIN
CATEGORY BASED DOCUMENT UPLOAD
SELECT CATEGORY
UPLOAD DOCUMENT
STORE DATABASE
OCR MATCH
GRAMMER, SPELLING MISTAKE ANALYZE
LLM PRETRAINED MODEL RECOMMENDATION
DOWNLOAD REPORT

3.3.2 DATA FLOW DIAGRAM

The Level 0 Data Flow Diagram shows the overall system as a single process that takes contract documents and organisational knowledge (policies and historical agreements) as input and produces risk analysis reports and clause recommendations as output. External entities are the Legal Analyst / Contract Manager and the organisational knowledge base.



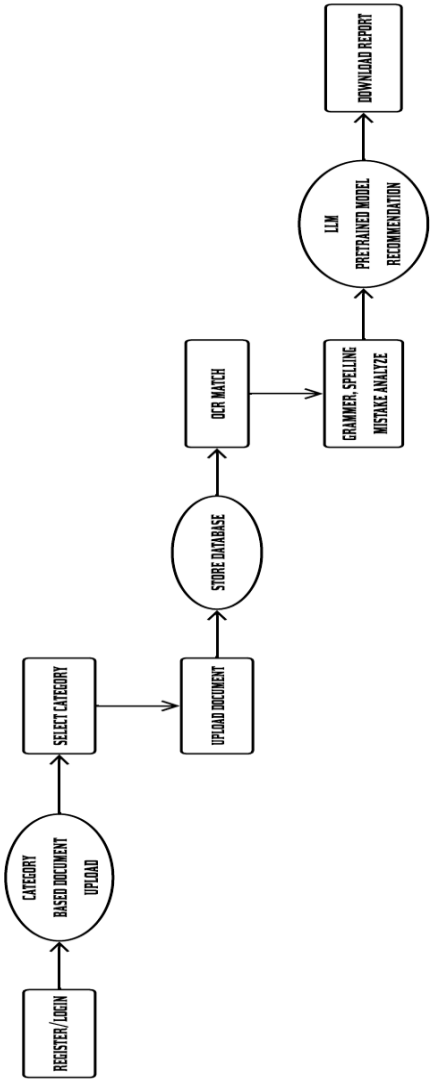
The Level 1 Data Flow Diagram expands this process into the individual modules: Document Processing, Clause Segmentation, Classification and Risk Scoring, Ollama Retrieval, Recommendation Generation, and Dashboard & Reporting. Data stores include the local storage of policies and historical agreements and a temporary store of analysed clauses with their risk scores and recommendations. Data flows between modules carry the extracted text, segmented clauses, classification labels, risk scores, retrieved context and generated recommendations.



The Level 3 Data Flow Diagram:



Overall Diagram:



4. METHODOLOGY

4.1 METHODOLOGIES

The development of the system follows an iterative and modular methodology. Each major component is designed, implemented and tested independently before being integrated into the complete pipeline. Transformer-based models are used for classification, dense vector embeddings support semantic retrieval, and large language models generate natural language recommendations. Continuous feedback from the dashboard interface allows refinement of risk scoring logic and recommendation quality.

The classification models are fine-tuned on the CUAD dataset, which provides expert annotations for 41 clause categories. Risk scoring combines the classification output with pattern matching against known high-risk formulations and, where available, similarity to previously labelled risky clauses. A locally hosted Large Language Model served through Ollama is prompted with the original clause, relevant organisational guidelines and surrounding contract context, together with explicit instructions to produce safer alternative wording while preserving commercial intent.

4.2 MODULES

4.2.1 DOCUMENT PROCESSING MODULE

This module accepts PDF and DOCX files. For text-based PDFs, pdfplumber extracts the content efficiently while preserving basic structure. For scanned documents, the pages are rendered as images and passed through Tesseract OCR. The extracted text is cleaned (removal of headers, footers, page numbers and excessive whitespace) and normalised before being passed to the next stage. The module also

records basic document metadata such as page count, file type and extraction method used.

4.2.2 CLAUSE SEGMENTATION MODULE

The cleaned text is segmented into individual clauses using a combination of rule-based heuristics and semantic boundary detection. Rule-based cues include numbered or lettered headings, section titles such as “Termination”, “Indemnity” or “Limitation of Liability”, and consistent indentation patterns. Semantic boundary detection helps identify clause breaks that are not marked by explicit headings. The goal is to produce coherent units of text that each represent a single contractual topic and are suitable for classification and risk scoring.

4.2.3 CLASSIFICATION AND RISK SCORING MODULE

Each segmented clause is passed to a fine-tuned LegalBERT, RoBERTa or DeBERTa model that assigns one or more legal category labels drawn from the CUAD taxonomy (or a simplified subset thereof). A subsequent risk scoring component evaluates the clause against known high-risk patterns (unlimited liability, automatic renewal, one-sided indemnity, ambiguous termination, etc.) and assigns a risk level of Low, Medium, High or Critical. The scoring logic can incorporate both the classification confidence and lexical or semantic similarity to previously labelled risky clauses.

4.2.4 OLLAMA RETRIEVAL MODULE

When a High or Critical risk clause is identified, the module prepares the original clause together with relevant organisational guidelines and surrounding contract context. This context is supplied to the locally hosted Ollama Large Language Model. Using a carefully designed

prompt, the model generates safer alternative wording that reflects the organisation's risk preferences while preserving the commercial intent of the original clause.

4.2.5 RECOMMENDATION MODULE

A locally hosted Large Language Model served through Ollama receives the original clause together with organisational guidelines and surrounding contract context, along with a carefully engineered prompt that instructs it to produce one or more improved alternative formulations. The prompt emphasises reduction of the identified risk, preservation of commercial intent, and adherence to organisational guidelines. The generated recommendations are returned so that a human reviewer can verify and accept or refine each suggestion.

4.2.6 DASHBOARD AND REPORTING MODULE

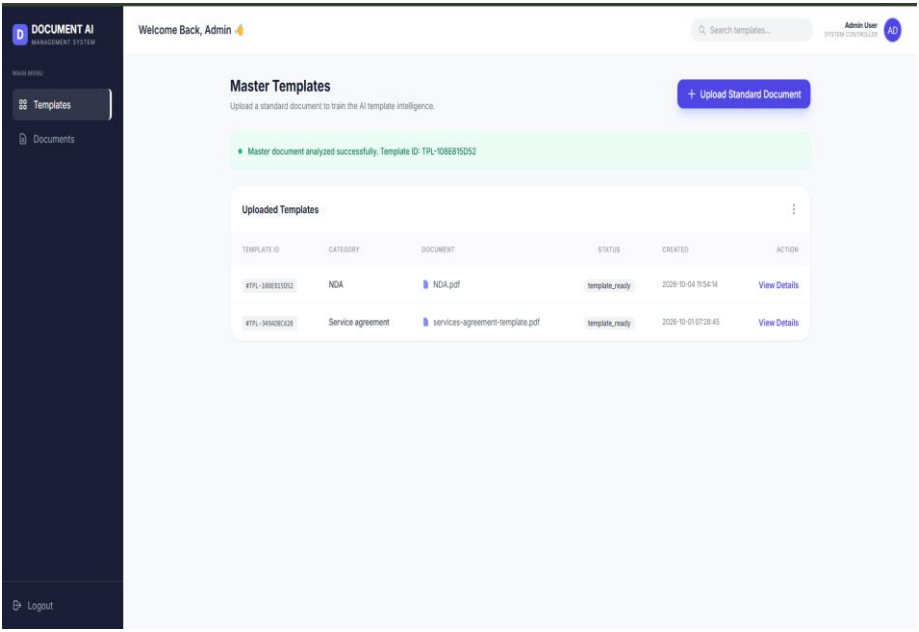
The results of the entire pipeline are presented through an interactive web dashboard built with Streamlit (or a similar framework). Users can view a visual summary of risk levels across the contract, drill down into individual clauses, examine the generated recommendations and the supporting retrieved evidence, and download a structured analysis report in PDF or Excel format. The dashboard is designed to support both quick triage of high-risk clauses and detailed review of specific recommendations.

5. IMPLEMENTATION RESULTS

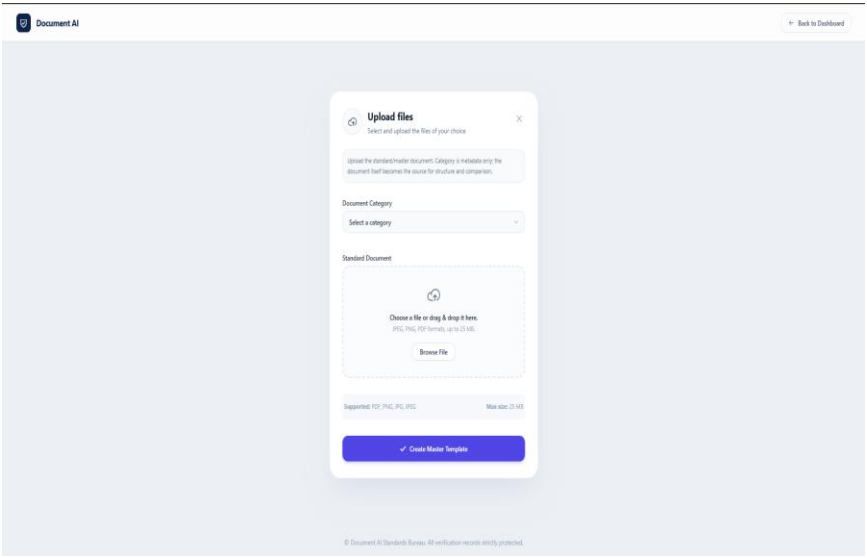
5.1 IMPLEMENTATION RESULTS

A working prototype of the system was developed and tested on a collection of sample commercial contracts. The prototype implements the core pipeline from document ingestion through classification, risk scoring, Ollama-based recommendation and recommendation generation, together with a basic interactive dashboard. The following subsections describe the outcomes observed for each major component. No invented quantitative performance figures are reported; the emphasis is on qualitative behaviour and the feasibility of the overall design.

Admin Page:



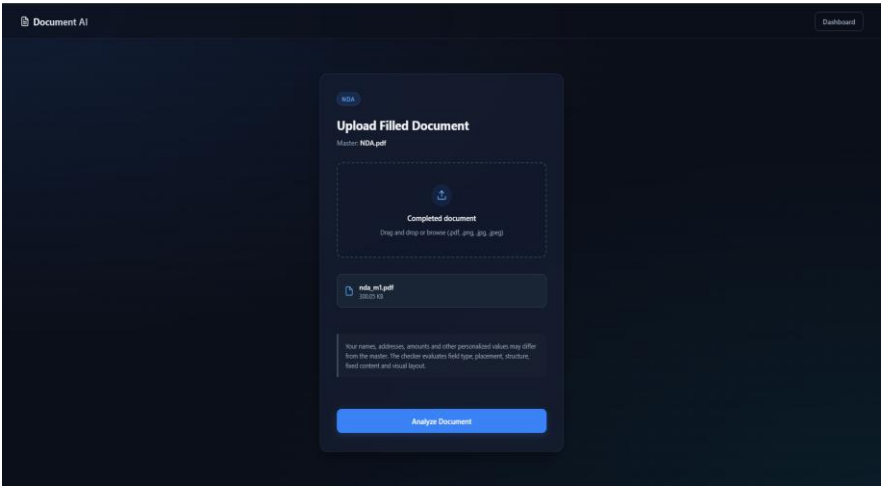
Uploading Templates:



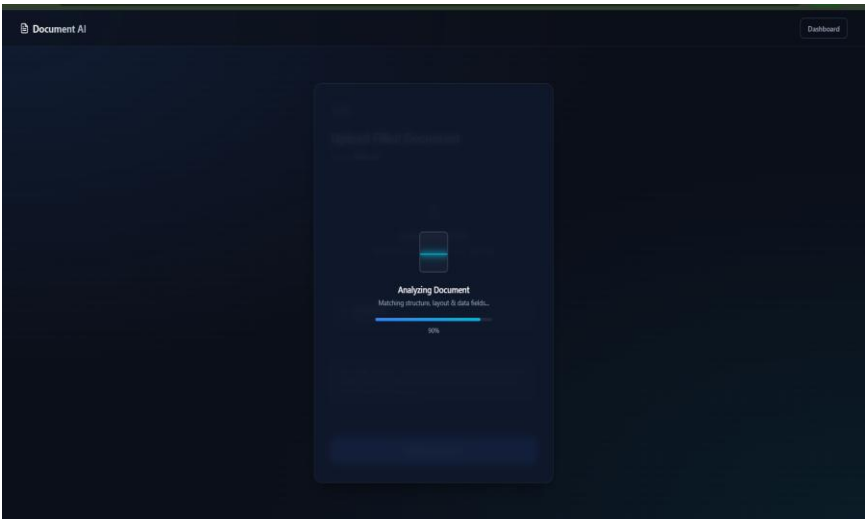
5.1.1 IMPLEMENTATION OF DOCUMENT PROCESSING

The document processing module successfully extracted text from both digitally created and scanned PDF contracts as well as DOCX files. For clean digital PDFs, pdfplumber produced accurate and well-structured text. For scanned pages, Tesseract OCR provided usable text with the expected residual recognition noise; subsequent cleaning steps removed most headers, footers and excessive whitespace. The module correctly recorded extraction metadata that is later displayed on the dashboard.

Document Upload:

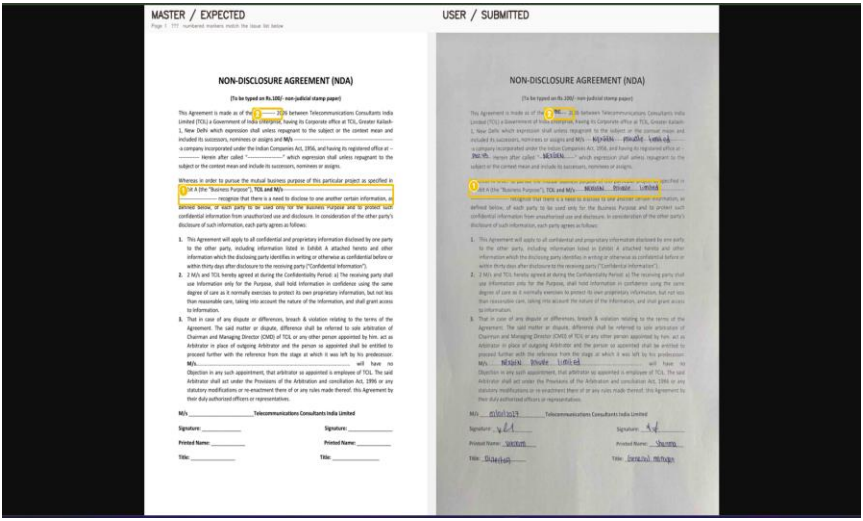


Document Processing:



5.1.2 IMPLEMENTATION OF CLAUSE CLASSIFICATION

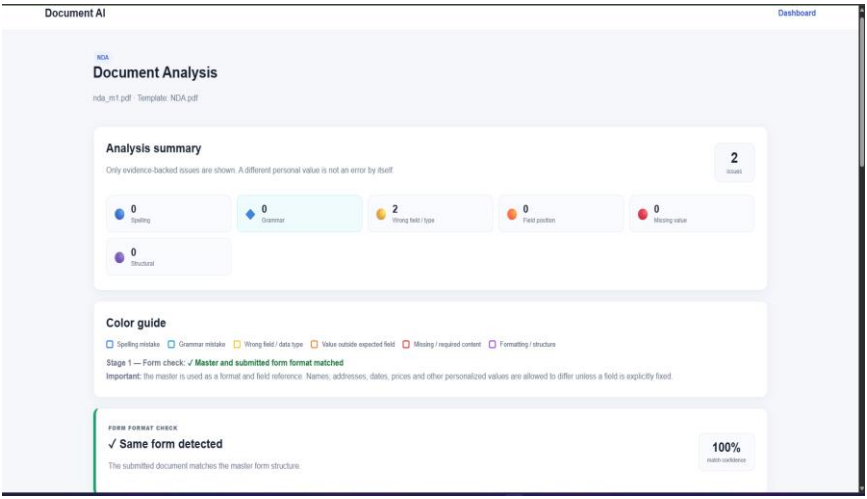
The classification component, based on a LegalBERT-style model fine-tuned on a subset of the CUAD categories, correctly identified common clause types such as liability, indemnity, termination, confidentiality, payment terms and governing law on the sample contracts. Clauses that did not match any of the fine-tuned categories were assigned a generic "Other" label. The prototype demonstrated that transformer-based classification is practical for the intended use case and can be extended to the full CUAD taxonomy in later phases.



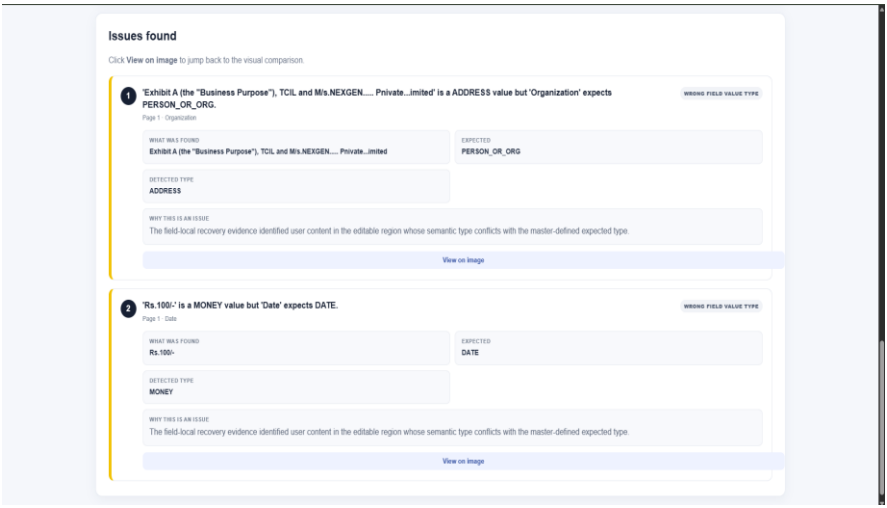
5.1.3 IMPLEMENTATION OF RISK SCORING

High-risk clauses, particularly those containing unlimited liability language, one-sided termination rights, automatic renewal with short notice windows, or vague and broad indemnity wording, were reliably flagged as High or Critical. Lower-risk administrative and boilerplate clauses received Low or Medium scores as expected. The four-level severity scale provided a clear prioritisation signal for subsequent human review and for the decision to invoke the Ollama and recommendation stages.

Analysis Summary:



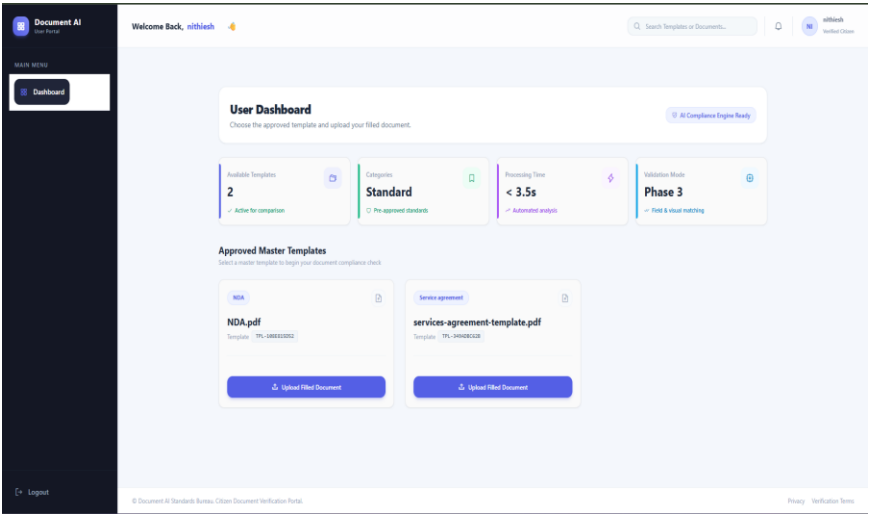
Issues Found:



5.1.4 IMPLEMENTATION OF DASHBOARD

The interactive dashboard provided a clear visual summary of risk levels across the contract and allowed users to navigate to individual clauses, view their classification and risk score, examine the generated recommendations and the supporting retrieved evidence, and export a structured report. The interface proved usable for both rapid triage of high-risk clauses and more detailed examination of specific recommendations. Further refinement of the visualisation and filtering capabilities is planned for subsequent phases.

User Page:



6. CONCLUSION

6.1 CONCLUSION

This Phase I report presents the complete design and initial prototype implementation of an AI-Based Contract Risk Analysis and Clause Recommendation System. The system successfully combines modern NLP techniques, local Large Language Model via Ollama and Large Language Models to automate substantial portions of the contract review process.

The literature survey confirmed the relevance and novelty of an end-to-end, organisation-adaptable approach that goes beyond pure classification to include risk scoring and grounded recommendation generation. System analysis established the clear need for such a solution given the limitations of both manual review and existing automated tools. The modular architecture provides a strong and extensible foundation for further development.

The prototype demonstrated the practical feasibility of the full pipeline—from document ingestion and clause segmentation through classification, risk scoring, Ollama-based recommendation and recommendation generation—together with an interactive dashboard for result presentation. Successful completion of Phase I has laid a solid base for building a more advanced, fully evaluated and organisation-deployable system in the subsequent phase.

6.2 FUTURE WORK

The following enhancements are planned for future work:

- Support for multiple languages beyond English, including the adaptation of classification models and OCR pipelines.
- Real-time collaborative review features that allow multiple users to annotate clauses, discuss recommendations and record decisions.
- Continuous learning from user feedback so that accepted or rejected recommendations improve future risk scoring and generation quality.
- Advanced explainability features, including confidence scores and more detailed evidence linking for every risk verdict and recommendation.
- Domain-specific fine-tuning for industries such as healthcare, information technology, construction and financial services.
- Exploration of multi-agent architectures for more complex legal reasoning, negotiation support and automated redlining workflows.
- Rigorous quantitative evaluation on held-out CUAD splits and on organisation-specific annotated contracts, together with user studies measuring review time and decision quality.

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